

Reading a Polynomial: Terms, Factors, Coefficients, Degree

Terms, coefficients and degree, with area models that turn products into sums and back

Directions

- Write every polynomial answer in standard form: highest power first, no like terms left.
 - Blanks inside a table or model are working space. The blank marked **Answer** is the one that is graded.
 - Show the step that got you there — a bare answer earns half credit.
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1. The table splits $P(x) = 4x^3 - 7x^2 + x - 9$ into its four terms. Fill in the missing coefficients and term degrees. Because $P(1)$ multiplies every power of x by 1, $P(1)$ is exactly the sum of the coefficients — use your completed table to find $P(1)$.

Term	Coefficient	Degree of the term
$4x^3$	_____	_____
$-7x^2$	_____	_____
x	_____	_____
-9	_____	_____

Answer: _____

2. An area model cuts a rectangle of length $(x + 6)$ and width $(x + 2)$ into four smaller rectangles. Fill in the four partial areas, then add them and write the total area in standard form.

$\times$	x	$+6$
x	_____	_____
$+2$	_____	_____

Answer: _____

3. Write $18x^3 + 30x^2$ as a product of its greatest common monomial factor and one remaining polynomial factor.

Answer: _____

4. Kian writes the polynomial as $3x(2x - 5)$. Nina writes it as $6x^2 - 15x$.

(a) Multiply out Kian's form to confirm the two expressions are equivalent.

Answer: _____

(b) One form shows the factors of the polynomial; the other shows its degree and its coefficients. Say which is which, and explain what each form makes easy to see.

Answer: _____

5. Multiply $(2x + 3)(x - 5)$ and write the result in standard form. State it so that the coefficient of each power of x is visible.

Answer: _____

- 6.** $Q(x) = (x + 4)(x - 1)(x + 2)$. The constant term of a polynomial is its value at $x = 0$. Use that fact to find the constant term of Q without multiplying the whole product out.

Answer: _____

- 7.** $A(x) = (3x^2 - 1)(x + 5)$ and $B(x) = 2x^4 - x^2$. Work out the degree of each one, then fill the blank with $<$, $>$, or $=$ so that the statement is true:

degree of A _____ degree of B

- 8.** Factor $x^2 - 3x - 28$ into two binomial factors.

Answer: _____

9. Consider $(x + 3)^2 - (x - 3)^2$.

(a) Expand both squares and simplify the difference.

Answer: _____

(b) Both starting pieces have degree 2, yet the answer does not. Explain what happened to the squared terms.

Answer: _____

10. A rectangle has area $6x^2 + 19x + 10$ and one side of length $(2x + 5)$.

(a) Find the length of the other side.

Answer: _____

(b) The area has degree 2 and each side has degree 1. Explain why the degrees of the two sides must add up to the degree of the area.

Answer: _____